

4/28/2026

Assignment no : 02

Design and Implementation of
Basic NAT (PAT) for Small
Network Connectivity

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Department : Cyber Security

Course : Computer Network

1. Objective

The objective of this assignment is to design and implement a basic NAT-enabled network using Cisco Packet Tracer. Specifically, the lab focuses on Port Address Translation (PAT) to allow multiple internal devices to communicate with an external network using a single public IP address.

2. Brief Explanation of NAT and PAT

- **NAT (Network Address Translation):** A method used to remap one IP address space into another by modifying network address information in the IP header of packets.
- **PAT (Port Address Translation):** Also known as NAT Overload, this is a type of dynamic NAT that maps multiple private IP addresses to a single public IP address by using different port numbers. This efficiently solves the problem of limited public IP addresses

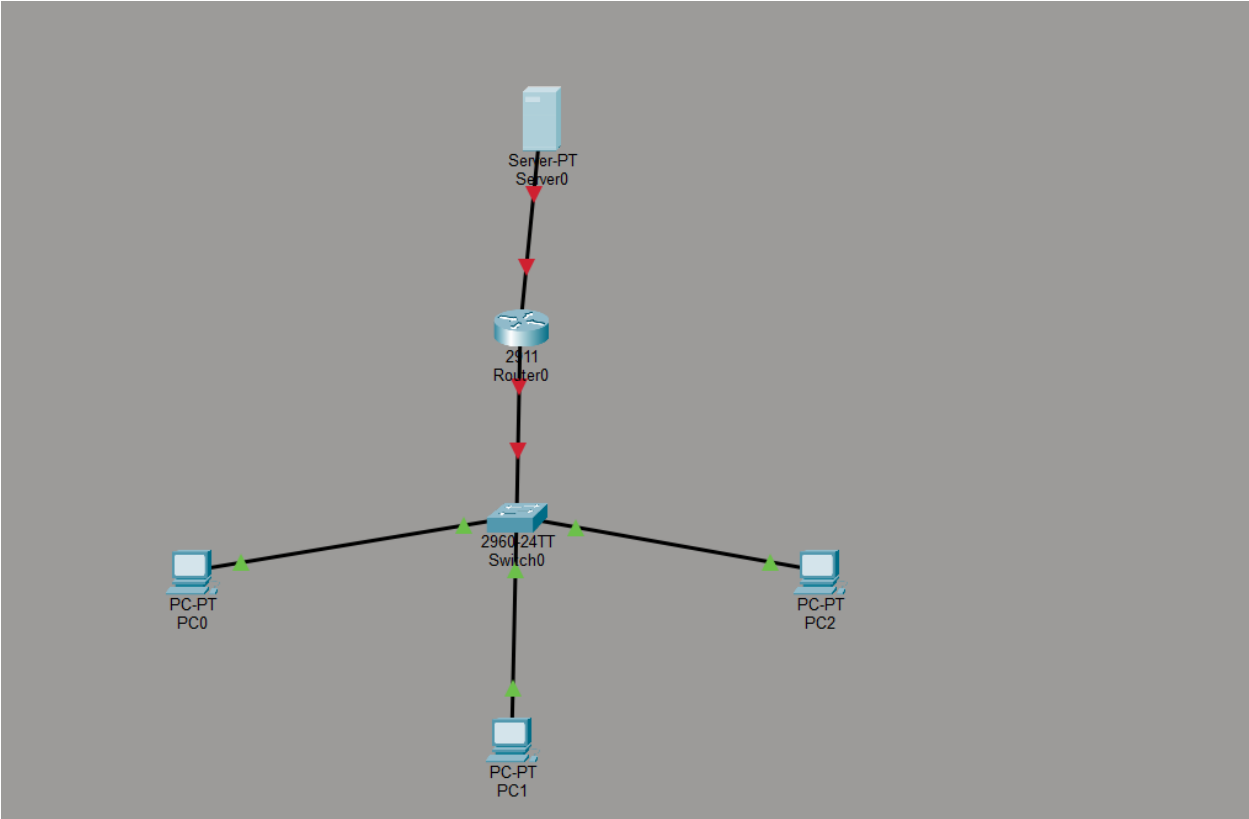
3. Packet Tracer Topology Setup:

You need a functional .pkt file with specific components:

- **Internal Network:** 3–5 PCs connected to a **Switch**.
- **Gateway:** A **Router** (e.g., 2911 model) connecting the Switch to the External Network.
- **External Network:** A **Server** representing the public internet.
- **Connections:** Copper straight-through cables for PC-to-Switch and Switch-to-Router; use a Serial or GigEthernet cable for Router-to-Server.

4. Network Topology Diagram

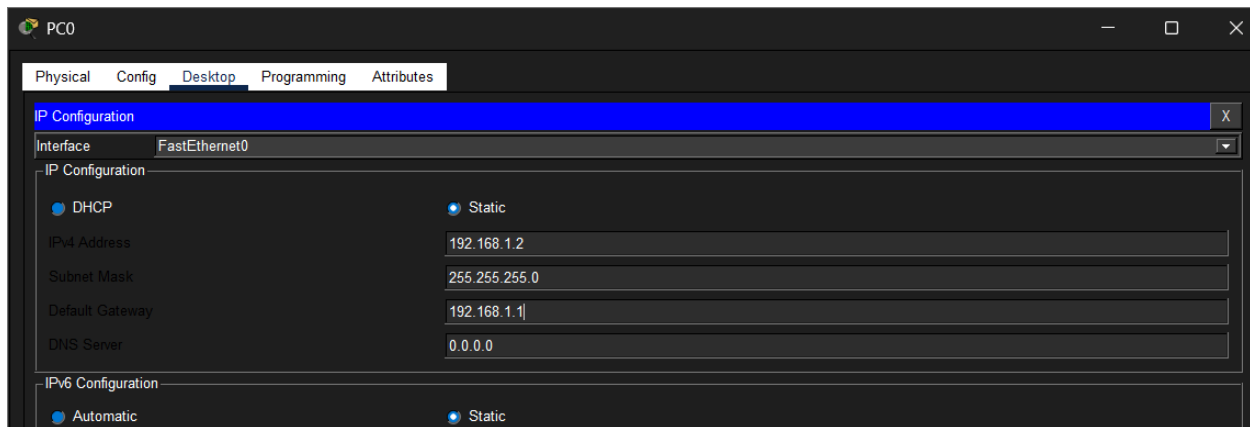
The network consists of a private segment with 3 PCs connected via a Switch to a Router, and a public segment containing a Server



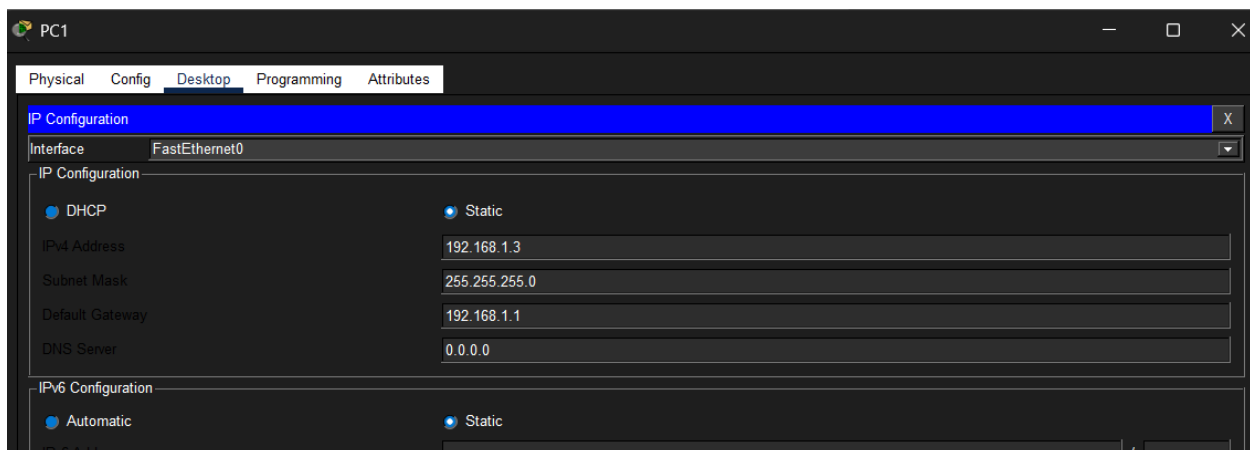
Device	Interface	IP Address	Role
Router	G0/0	192.168.1.1/24	Inside (Gateway)
Router	G0/1	200.1.1.1/24	Outside
PC1	NIC	192.168.1.2/24	Internal
PC2	NIC	192.168.1.3/24	Internal

Device	Interface	IP Address	Role
Server	NIC	200.1.1.2/24	External

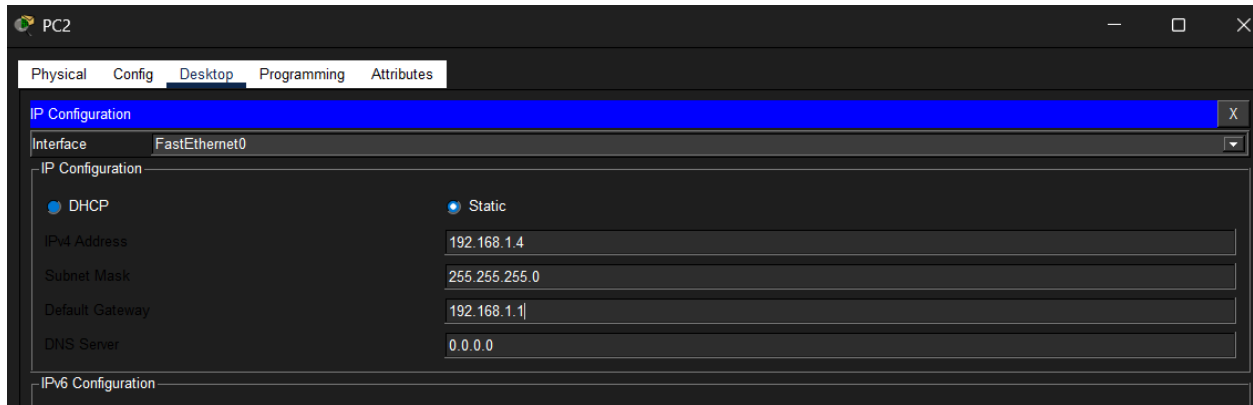
Assigning IP to Pc 0 :



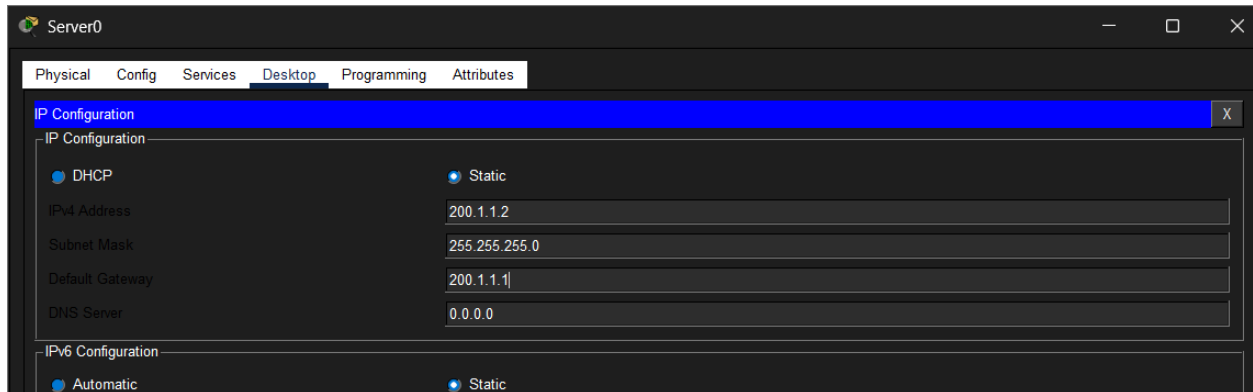
Assigning IP to Pc 1 :



Assigning IP to Pc 2:



Assigning IP to Server:



Configuration Commands For Router :

```

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
  
```

Assigning IPs to Router interface:

```
Vlan1          unassigned      YES unset  administratively down down
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#shoow ip interface brief
      ^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  200.1.1.1      YES manual up          up
GigabitEthernet0/2  unassigned     YES unset  administratively down down
Vlan1           unassigned     YES unset  administratively down down
Router#
```

```
% Invalid input detected at '^' marker.
```

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
```

Ping from PC0 to Server0:

```
C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time=1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

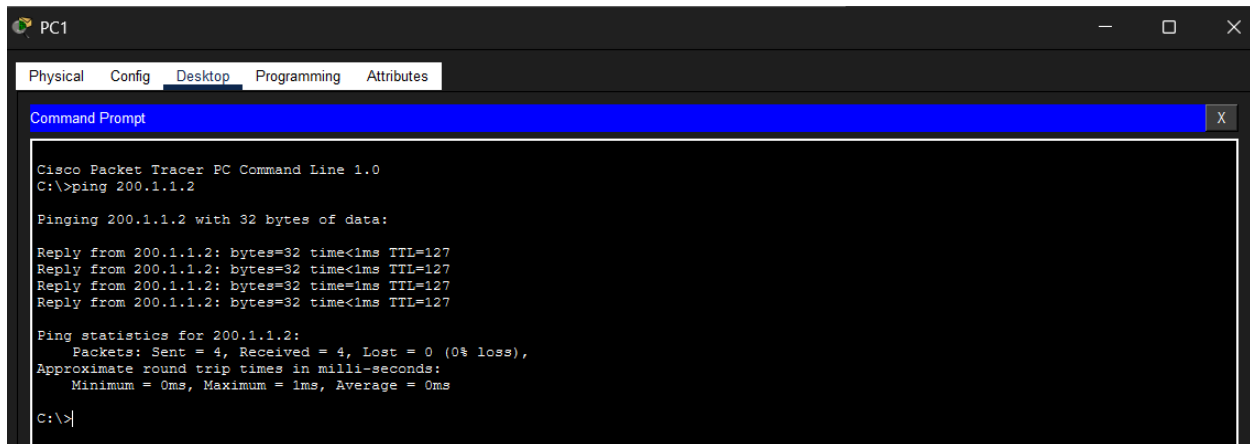
C:\>
```

Top

NAT Translation Table:

```
Router>enable
Router#show ip nat translation
Router#show ip nat translations
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.1:1        192.168.1.2:1     200.1.1.2:1        200.1.1.2:1
icmp 200.1.1.1:2        192.168.1.2:2     200.1.1.2:2        200.1.1.2:2
icmp 200.1.1.1:3        192.168.1.2:3     200.1.1.2:3        200.1.1.2:3
icmp 200.1.1.1:4        192.168.1.2:4     200.1.1.2:4        200.1.1.2:4
Router#
```

Ping from PC1 to Server0:



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC1. The command prompt displays the output of a ping command to 200.1.1.2. The output shows four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 127. The ping statistics show 4 packets sent, 4 received, and 0% loss.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

NAT Translation Table:

```
icmp 200.1.1.1:4        192.168.1.2:4     200.1.1.2:4        200.1.1.2:4
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.1:1        192.168.1.3:1     200.1.1.2:1        200.1.1.2:1
icmp 200.1.1.1:2        192.168.1.3:2     200.1.1.2:2        200.1.1.2:2
icmp 200.1.1.1:3        192.168.1.3:3     200.1.1.2:3        200.1.1.2:3
icmp 200.1.1.1:4        192.168.1.3:4     200.1.1.2:4        200.1.1.2:4
Router#
```

Copy

Paste

Ping from PC0 to PC1:

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Configuration Command:

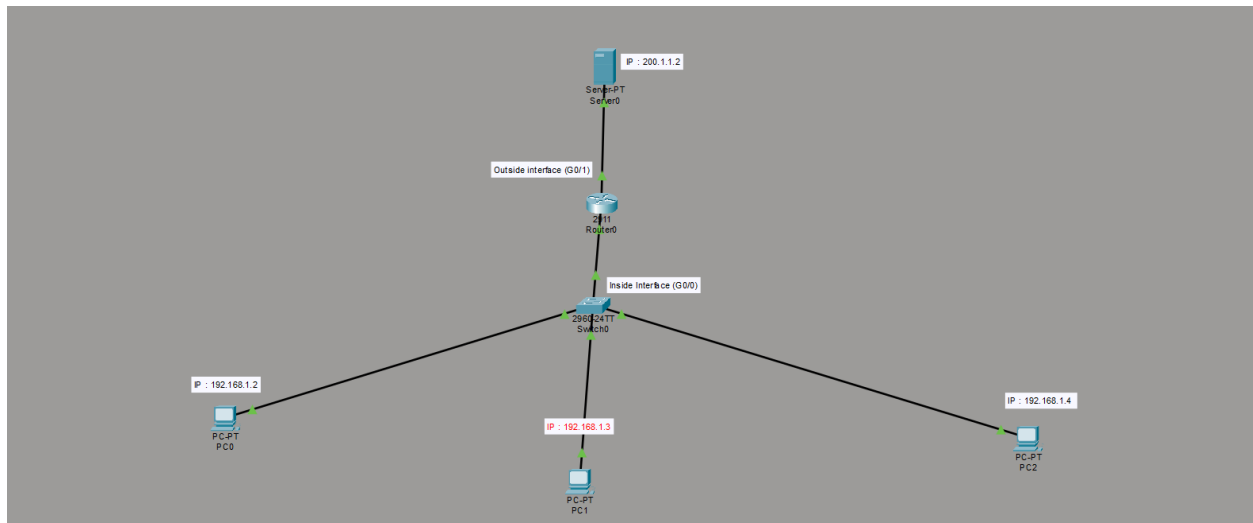
```
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show running-config
Building configuration...

Current configuration : 857 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
--More--
```

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Final Result:



Conclusion

The implementation was successful. The results demonstrate that PAT effectively allows a small network to share a single public IP for external connectivity while maintaining individual communication sessions through port differentiation.